

A Data-Fitted Effective-Analyzer Model for ϕ_2 -Dependent Output Amplitude in a Hybrid Mach–Zehnder Interferometer

Experiment analysis notes

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Abstract

This note derives and audits a data-fitted prediction of the ϕ_2 -dependent power modulation measured at the two final outputs of a hybrid Mach–Zehnder interferometer. A two-axis A-only/B-only/both-path calibration fits an effective Stokes analyzer for each instrument. A later high-contrast both-path fringe map supplies the current ϕ_1 branch. The conditional transfer calculation predicts mean ϕ_2 fringe contrast 0.571 at the photodiode and 0.619 in PAX `ptotal` units; the corresponding measured means are 0.585 and 0.657. The contrast mean absolute errors are 0.027 and 0.041. This is evidence that the fitted model predicts modulation *magnitude*; it is not a blind prediction of fringe phase.

1 Evidence and data sets

The model coefficients came from `192530_field-model-calibration_phi2-power-model-0/data.csv`. That file contains 984 simultaneous PAX/PD samples: four ϕ_1 RP biases, 41 ϕ_2 commands spanning the assumed V_λ , three blocking conditions (path A only, path B only, and both paths), and forward/reverse passes. Its SHA-256 prefix is `70aa7cfe62be8823`.

The transfer target is `200934_phi1-fringe-map_phi1-fringe-map-1/data.csv`, containing 714 simultaneous PAX/PD samples with both paths open: 17 ϕ_1 commands, 21 ϕ_2 commands per fringe, and forward/reverse ϕ_1 passes. Its SHA-256 prefix is `e8719481ab444339`.

The saved coefficient and comparison files are `effective-analyzer-fit.csv` and `phi2-amplitude-comparison.csv`; their SHA-256 prefixes are `3c803a9e5ea4ecad` and `b59266a8c88eb988`, respectively. These source files are retained in the experiment folders beside this paper.

2 From ideal propagation to a detector-power model

For the ideal equal-amplitude hybrid-MZI reference at the selected output port, the PAX-convention Stokes vector is

$$\mathbf{S}(\alpha, \phi_2) = \begin{bmatrix} \cos \phi_2 \\ \sin \phi_2 \cos \alpha \\ \sin \phi_2 \sin \alpha \end{bmatrix}, \quad \alpha = \phi_1 + \delta \text{ in the ideal static limit.} \quad (1)$$

The observed ϕ_1 response is history dependent, so the analysis does not force α to equal a linear RP-voltage scale. Instead, α is an empirical branch coordinate.

The smallest model identifiable from the present two-axis data is a detector response linear in Stokes state:

$$P_d = B_d + \mathbf{h}_d \cdot \mathbf{S} \quad (2)$$

$$= B_d + h_{1,d} \cos \phi_2 + [h_{2,d} \cos \alpha + h_{3,d} \sin \alpha] \sin \phi_2. \quad (3)$$

Here $d \in \{\text{PD}, \text{PAX}\}$, while B_d and $\mathbf{h}_d$ are in the native units of that instrument. Equation (3) is an *effective* analyzer model: it parameterizes the measured power response, but does not by itself locate the particular nonideal optical component responsible.

At fixed α , define

$$C_d = h_{1,d}, \quad D_d = h_{2,d} \cos \alpha + h_{3,d} \sin \alpha. \quad (4)$$

Then $P_d = B_d + C_d \cos \phi_2 + D_d \sin \phi_2$, and the predicted amplitude and contrast are

$$A_{d,\text{pred}} = \sqrt{C_d^2 + D_d^2}, \quad \mathcal{C}_{d,\text{pred}} = \frac{A_{d,\text{pred}}}{B_d}. \quad (5)$$

The photodiode OD-2 attenuation changes its absolute voltage but cancels from $\mathcal{C}$; PD volts and PAX **ptotal** units are never treated as an absolute common power scale.

3 Calibration coefficients

For each detector and path condition, least squares fits the design row

$$\mathbf{x} = (1, \cos \phi_2, \sin \phi_2 \cos \phi_1, \sin \phi_2 \sin \phi_1) \quad (6)$$

to obtain $(B_d, h_{1,d}, h_{2,d}, h_{3,d})$. The both-path coefficients used in the later transfer calculation are listed in Table 1.

Table 1: Both-path effective-analyzer coefficients from the calibration.

Detector	B	h_1	h_2	h_3	R^2	RMS
PD (V)	0.2433345	0.1320399	0.0105049	0.0429494	0.900	0.03223
PAX (ptotal)	2.311569×10^{-4}	-1.326148×10^{-4}	-9.59709×10^{-6}	-5.35634×10^{-5}	0.793	5.04244×10^{-5}

The nonzero $\mathbf{h}_d$ values quantify the observed phase-to-power dependence. In the ideal constant-total-power result, a true total-power measurement would have no Stokes-dependent term.

4 Measured fringe contrast and empirical branch selection

At every fixed ϕ_1 command and scan direction in the later map, each detector is independently fit to

$$P_{d,\text{meas}} = b_d + c_d \cos \phi_2 + d_d \sin \phi_2. \quad (7)$$

The measured contrast and phase are

$$\mathcal{C}_{d,\text{meas}} = \frac{\sqrt{c_d^2 + d_d^2}}{b_d}, \quad \psi_d = \text{atan2}(-d_d, c_d). \quad (8)$$

The current phase branch is selected, independently at each ϕ_1 command and direction, by a dense numerical minimization over $\alpha \in [0, 2\pi)$:

$$\alpha_* = \underset{\alpha}{\text{argmin}} \sum_d \left[\frac{h_{2,d} \cos \alpha + h_{3,d} \sin \alpha}{B_d} - \frac{d_d}{b_d} \right]^2. \quad (9)$$

Equation (9) uses the later map's *sine quadrature*, i.e. fringe-phase information. It does *not* use its measured contrast as an input. Once α_* is selected, Eq. (5) produces the predicted amplitude/contrast.

This distinction is essential: the result is a conditional transfer validation of amplitude magnitude, not a blind future-state prediction of both phase and amplitude.

5 Worked calculation at forward $\phi_1 = 0$ RP V

For this point, joint PD/PAX branch selection gives $\alpha_* = 0.032044$ rad. For the PD,

$$\frac{C_{\text{PD}}}{B_{\text{PD}}} = \frac{h_{1,\text{PD}}}{B_{\text{PD}}} = 0.5426272, \quad (10)$$

$$\frac{D_{\text{PD}}}{B_{\text{PD}}} = \frac{h_{2,\text{PD}} \cos \alpha_* + h_{3,\text{PD}} \sin \alpha_*}{B_{\text{PD}}} = 0.0488033, \quad (11)$$

$$\mathcal{C}_{\text{PD,pred}} = \sqrt{(C_{\text{PD}}/B_{\text{PD}})^2 + (D_{\text{PD}}/B_{\text{PD}})^2} = 0.5448174, \quad (12)$$

$$\mathcal{C}_{\text{PD,meas}} = 0.5457103. \quad (13)$$

For PAX, $(C/B, D/B) = (-0.5737002, -0.0489203)$, which gives

$$C_{\text{PAX,pred}} = 0.5757821, \quad C_{\text{PAX,meas}} = 0.5878357. \quad (14)$$

6 Aggregate comparison

All 17 ϕ_1 commands and both directions are included ($N = 34$ per detector). Mean absolute error is defined by

$$\text{MAE}_d = \frac{1}{N} \sum_{i=1}^N \left| C_{d,\text{pred}}^{(i)} - C_{d,\text{meas}}^{(i)} \right|. \quad (15)$$

Table 2: Predicted and measured ϕ_2 fringe contrasts.

Detector	mean pred.	mean meas.	MAE	RMSE	bias	median relative error
PD	0.571	0.585	0.027	0.034	-0.014	4.3%
PAX	0.619	0.657	0.041	0.048	-0.038	6.1%

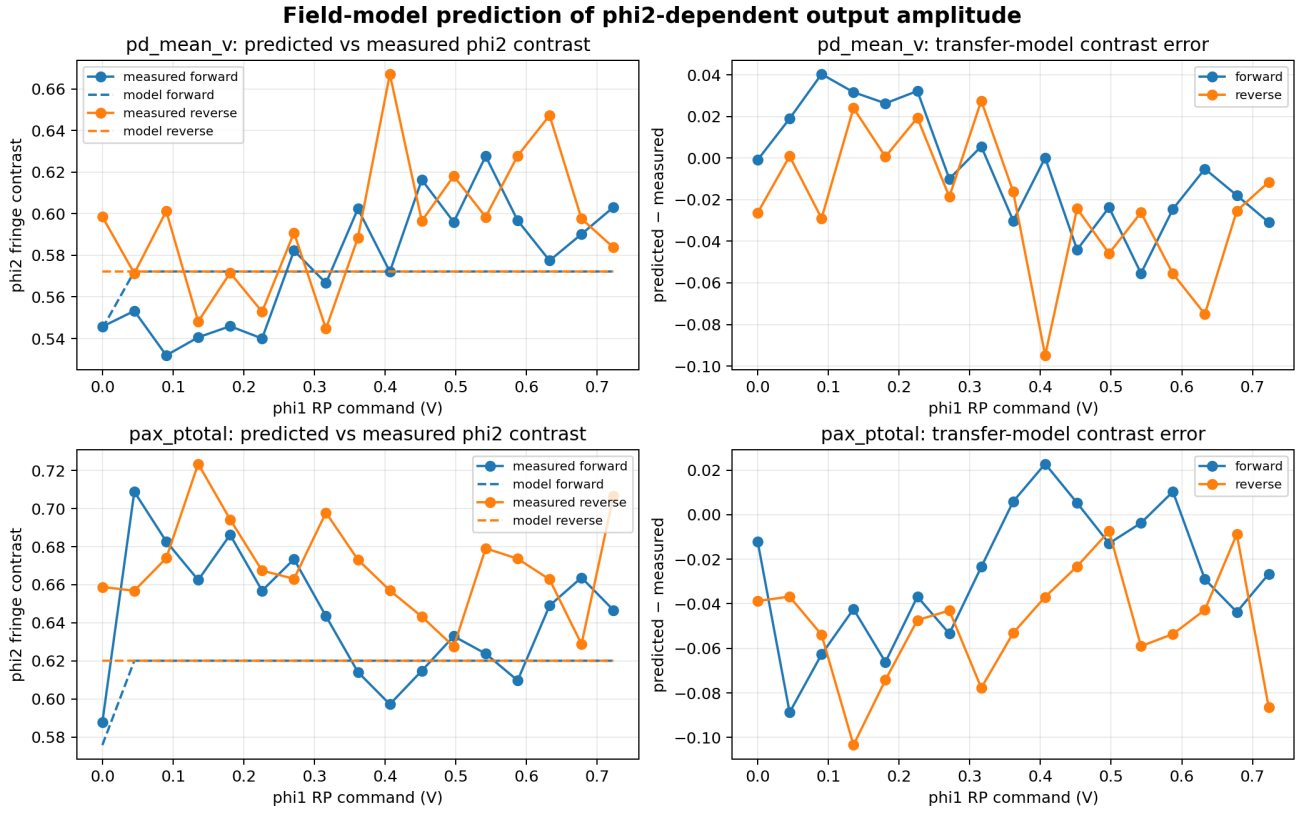


Figure 1: Stored numerical comparison. Dashed traces are the conditional effective-analyzer prediction; points are measured fringe contrasts.

7 Interpretation and next experiment

The calculation quantitatively captures the observed *degree* of ϕ_2 -dependent output modulation in both instruments. It therefore supports a real, substantial phase-to-power term beyond the ideal constant-power model. It does not uniquely distinguish nonideal beam splitting, polarization leakage, mode overlap, path-dependent loss, or detector response; multiple physical mechanisms can produce the same effective $\mathbf{h}_d$.

The most important limitation is that phase/quadrature transfer is not yet clean and α was conditioned on the later map. The next decisive experiment is a high-visibility **field-model-calibration** followed immediately by **phi1-fringe-map**. Repeating the present calculation on that state-matched pair will test whether the same model predicts amplitude without the present run-to-run analyzer/branch mismatch.